

Institute of Actuaries of India

Subject SP6 – Financial Derivatives Principles

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

- i) First calculate the price of the zero-coupon bonds from times 3 to 3.5, 3.5 to 4.0, ..., 5.5 to 6.0. For this, one can use the formula $ZCB(t) = \exp(-t * i_t)$ using the data given in the question. Thus, one would get

Term (years)	Risk Free Spot Rate	ZCB Price
3.0	1.15%	0.96609
3.5	1.32%	0.95485
4.0	1.48%	0.94252
4.5	1.64%	0.92886
5.0	1.80%	0.91393
5.5	1.94%	0.89880
6.0	2.08%	0.88267

Then the forward swap rate is $2 * (ZCB_3 - ZCB_6) / (ZCB_{3.5} + \dots + ZCB_6) = 3.02\%$

[3]

- ii) Value of a stream of payments of 0.5 at times 3.5, ..., 6 is

$$A = 0.5 * (ZCB_{3.5} + \dots + ZCB_6) = 2.76081$$

Value of the FRA is then Notional * A * (Fixed rate – Swap rate)

$$= 10,00,00,000 * 2.76081 * (3.5\% - 3.02\%) = \text{Rs. } 13,21,000$$

Thus, the financial institution is the net receiver and the amount is Rs.13,21,000.

[3]

- iii) Price of a receiver swaption is $LA[s_K N(-d_2) - s_0 N(-d_1)]$

Where

- $d_1 = [\ln(s_0/s_K) + \sigma^2 T/2] / \sigma \sqrt{T}$
- $d_2 = d_1 - \sigma \sqrt{T}$
- σ is Black volatility = 0.2
- s_0 is forward swap rate = 0.0302
- s_K is strike (at-the-money) = 0.0302
- L is notional = Rs. 10 crores
- A is as above: value of a stream of payments of 0.5 at times 3.5, ..., 6 = 2.76081
- T = 3 years

In this case:

- $d_1 = 0.17321 \Rightarrow N(-d_1) = 0.43125$
- $d_2 = -0.17321 \Rightarrow N(-d_2) = 0.56875$

So, the swaption price is

$$\text{Rs. } (100,000,000 * 2.76081 * 3.02\% * (0.56875 - 0.43125)) = \text{Rs. } 11,47,000$$

[5]

- iv) The purchase of the FRA hedges the financial institution's market risk perfectly.

The swaption, on the other hand:

- Allows the financial institution to profit from interest rate increases: if the three-year swap rate at time 3 exceeds 3.5%, then the firm benefits from earning a higher rate on the Rs. 10

crores than it is paying out (with the FRA, the firm would be hedging away profits as well as losses).

- Under-protects the firm by having a strike rate of 3.02% rather than the desired 3.5%. If the three – year swap rate at time $t = 3$ is below 3.02%, the firm will lose 0.48% on a net basis, this being the difference between the 3.02% received and the 3.5% paid. If the three – year swap rate at time 3 is between 3.02% and 3.5%, the swaption expires worthless and the firm is left to pay out 3.5%, which exceeds the swap rate at that time.

As a result of the value of the under-hedging exceeding the value of the extra profits in the “interest rate up” scenarios, the FRA is more expensive than the swaption.

The firm needs to also consider the credit risk associated with the two possible hedges. Today the credit exposure is at a similar level for the two hedges. If interest rates increase above 3.5%, the FRA will become a liability to the financial institution (so not a credit risk) and the swaption would have a small time value.

If interest rates remain below 3.5%, then close to maturity the FRA would be worth more than the swaption so represents a bigger credit risk, and further from maturity the swaption will have some time value, so things will be less clear.

It is also worth noting that the FRA (but not the swaption) can become a liability to the financial institution and an asset to the counterparty. If the counterparty has insisted on credit risk mitigation processes being in place, the financial institution could find itself being called upon to post collateral with the counterparty.

The swaption will also expose the financial institution to Vega risk.

[5]

[16 Marks]

Solution 2:

- i) The value of the trader's position is $V_t = \phi_t S_t - C_t$. V_t is risk free if and only if the coefficient of the randomness dW_t is zero, during the small-time interval dt . The change in the value during dt is

$$dV_t = \phi_t dS_t - dC_t = \phi_t (\mu S_t dt + \sigma S_t dW_t) - (\mu_t^C C_t dt + \sigma_t^C C_t dW_t).$$

Since the trader does not change; it is therefore risk-free if and only if

$$\phi_t \sigma S_t = \sigma_t^C C_t.$$

[3]

- ii) The condition of no arbitrage means that the drift in the no – risk position must be the short-term interest rate. Thus

$$\phi_t \mu S_t - \mu_t^C C_t = r V_t = r(\phi_t S_t - C_t),$$

which, on substituting ϕ_t gives

$$\sigma_t^C \frac{C_t \mu}{\sigma} - \mu_t^C C_t = r \left(\frac{\sigma_t^C C_t}{\sigma} - C_t \right).$$

Dividing both sides by $C_t \sigma_t^C$ gives the result. This means that there exists a common risk premium $\lambda = (\mu - r)/\sigma$, the same for all the assets and all derivatives of it.

[3]

iii) By Ito's Lemma

$$dC_t = \frac{\partial C}{\partial t} dt + \frac{\partial C}{\partial S} dS + \frac{1}{2} \frac{\partial^2 C}{\partial S^2} (dS)^2$$

as the other second derivatives are zero by the Ito rules. The only non-zero term is $(dS)^2$ is $\sigma^2 S_t^2 (dW_t)^2 = \sigma^2 S^2 dt$. Therefore,

$$dC = \left(\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S} \mu S + \frac{1}{2} \frac{\partial^2 C}{\partial S^2} \sigma^2 S^2 \right) dt + \sigma S \frac{\partial C}{\partial S} dW_t.$$

We then prove part c by identifying $C_t \mu_t^C$ with the coefficient of dt and $C_t \sigma_t^C$ with the coefficient of dW_t .

[4]

iv) Using the earlier result from part b, we have

$$\mu_t^C = r + \sigma_t^C \frac{\mu - r}{\sigma} = r + \left(\frac{S}{C} \right) \frac{\partial C}{\partial S} (\mu - r).$$

Substituting in part c, we get

$$\mu^C = C^{-1} \left(\frac{\partial C}{\partial t} + \mu S \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} \right) = r + \left(\frac{S}{C} \right) \frac{\partial C}{\partial S} (\mu - r).$$

Thus, the result follows.

This means that the derivatives can be priced without knowing the rate of return on the asset.

[4]

[14 Marks]

Solution 3:

- i) Barrier options are options where the payoff depends on whether the underlying asset's price reaches a certain level during a certain period of time; in essence it is a path dependant option.

Single barrier options can be classified as either knock-out options or knock-in options.

A knock-out option ceases to exist when the underlying asset price reaches a certain barrier.

A knock-in option comes into existence only when the underlying asset price reaches a barrier.

Single barrier options come in four different types: up & in, up & out, down & in, down & out.

For "down" barriers the spot price starts above the barrier level and must move down for an event to occur. For "up" barriers the spot price starts below the barrier level and must move up for an event to occur.

Barriers can also come in other forms: double barriers, Parisian options, fluffy barriers, partial timed barrier options, window barriers and several more.

Barrier options can be calls or puts and can be exercised as American, European or Bermudan options.

[1]

- ii) Barrier option payoffs may more closely match views about the future behaviour of the market. For example, an investor buying a barrier option can eliminate paying for scenarios which are deemed unlikely by the investor. Or, alternatively an investor selling a barrier option can enhance the return by selling a barrier option which only pays on scenarios which are deemed improbable.

Barrier options may be more appropriate for hedging particular exposures than other options, because the pay-outs are limited to particular scenarios.

The premium of Barrier options are generally lower than those of similar vanilla options. This is because when a barrier option is compared to an equivalent vanilla option with the same strike price, period to expiry and the same underlying, it is identical apart from paying out in fewer scenarios.

[1]

- iii) Consider a portfolio of one up & in single barrier option and one up & out single barrier option. Both of these should be European call in nature having a strike and barrier equal to the strike price of the European vanilla call option to be replicated.

At expiry just one of the barrier options will be active as the other will be worthless due to the barrier being crossed. As the payoff from either option when it is active is identical to a European call option, it follows that at expiry the payoff from the portfolio will be identical to the European call option.

If the option starts out of the money, only the up & in single barrier option with strike price = barrier price is needed.

[1]

iv)

- a) If the barrier is not hit during the lifetime then the pay-out is the same as for the vanilla put option, i.e. the pay-out is the strike price less the price of the underlying asset if this is greater than zero.

It should be noted that the payoff is limited due to the barrier; unlike with the standard put option which has an unlimited pay-out.

If the barrier is hit (i.e. the price falls below the barrier level at any time during the year); then the underlying option is knocked out and there is no pay-out.

[2]

- b) The graph of the delta of the barrier option can be segmented into four regions.
- When the exchange rate is much larger than the barrier of 0.7, the impact of the barrier is negligible. Therefore in this region the delta is negative like that of a vanilla put option. Asymptotically, the delta will tend towards zero from below.
 - When the exchange rate is lower than the barrier the barrier option is knocked-out. The delta in this region is therefore zero.
 - In the region just above the barrier, a small increase in the exchange rate will reduce the probability of the barrier option being knocked-out. This would therefore lead to a large increase in the barrier option's value. Hence, the delta of the barrier option will be large and positive.
 - At the barrier itself, there is a discontinuity in the payoff. The delta becomes infinitely large.

Overall, the fluctuation of the barrier option's delta from positive to negative is as a result of a decrease in the value of the barrier option but with increasing probability of it not being knocked out.

[3]

- c) The bank is delta hedging a long position with nominal 20 million AUD. It therefore needs to sell 20 million times the delta of the barrier option in AUD.

If the spot price approaches the barrier then the delta becomes larger. This requires the bank to sell more AUD to maintain the delta hedge. This can influence the currency market, as gradually selling AUD will accelerate the movement of the spot price towards the barrier (in isolation from other market influences).

In extreme cases this could lead to the barrier being breached.

A more likely scenario is that the bank will have insufficient capacity to continue purchasing AUD.

If the option is knocked-out then there is no barrier option to hedge. As a result of the option being knocked-out, there may be less or no downward movement as fewer AUD are being sold in the market. In extreme cases the bank may need to unwind the hedge quickly, resulting in a large upward movement in the spot price.

[2]

[10 Marks]**Solution 4:**

- i) The Vasicek model and its statistical properties:

- This is a one-factor model because it incorporates a single source of randomness and is used for modelling the short rate of interest $r(t)$.
- It assumes that $r(t)$ has the dynamics of an Ito process (in fact, an Ornstein – Uhlenbeck process) under the risk-neutral probability measure and is expressed as $dr(t) = \alpha[\mu - r(t)]dt + \sigma dW(t)$, where $W(t)$ is standard Brownian motion
- The movements in the interest rate are therefore normally distributed (and hence can take negative values).
- The parameter σ determines the volatility.
- The parameter α is assigned a positive value, so that $r(t)$ is stationary and mean reverts to the constant value μ . A higher value of α implies more rapid mean reversion.

[2]

ii)

a)

We can rewrite the given function as:

$$f(t) = r_0 e^{-\alpha t} + r_\infty (1 - e^{-\alpha t}) + k(1 - e^{-\alpha t}) e^{-\alpha t} = r_\infty + (r_0 - r_\infty + k) e^{-\alpha t} - k e^{-2\alpha t}$$

Differentiating,

$$f'(t) = -(r_0 - r_\infty + k) \alpha e^{-\alpha t} + 2k \alpha e^{-2\alpha t} = [- (r_0 - r_\infty + k) + 2k e^{-\alpha t}] \alpha e^{-\alpha t}$$

For a stationary point, this must equal zero and so the expression in square brackets must also equal zero. Thus

$$-(r_0 - r_\infty + k) + 2k e^{-\alpha t} = 0 \Rightarrow e^{-\alpha t} = \frac{r_0 - r_\infty + k}{2k}$$

[2]

- b) Graph (1) will have a stationary point if

$$e^{-\alpha t} = \frac{0.04 - 0.06 + 0.1}{2(0.1)} = 0.4$$

There will be one solution to this equation, corresponding to a positive value of τ .

Also, when $\tau = 0$, the slope is

$$f'(0) = [-(r_0 - r_\infty + k) + 2k]\alpha = [k - (r_0 - r_\infty)]\alpha = [0.1 - (0.04 - 0.06)]\alpha > 0$$

So graph (1) will be a humped graph starting at 0.04 (4%) for the short rate and ending at 0.06 (6%) for the long rate.

Graph (2) will have the stationary point if

$$e^{-\alpha t} = \frac{0.06 - 0.04 + 0.02}{2(0.02)} = 1$$

So this will have a stationary point at $\tau = 0$ only.

So Graph (2) will start at 0.06 (6%) and then fall steadily to 0.04 (4%).

[2]

- c) The usual explanation for a humped yield curve is market segmentation. Different categories of investors are attracted to bonds with different terms (short, medium or long). The bond prices, and hence the yields within each “segment” of the market, are determined by demand and supply.

In fact, graph (1) is a common shape for the yield curve.

[2]

iii) Advantages and limitations

- The mean-reverting property is consistent with real-world observations.
- The model is mathematically tractable, with few parameters to estimate.
- The statistical properties are based on the familiar normal distribution.
- A variety of shapes can be obtained for the yield curve, *i.e.* upward-sloping, downward sloping and humped.
- The model leads to arbitrage-free yield curves.
- The model has the disadvantage that it allows $r(t)$ to take negative values, which we would not expect to observe in the real world. While this has generally been true, however, in the current environment, we have also seen negative interest rates in a number of economies and in that context, the jury is probably still out as to whether negative interest rates are sustainable.

[2]

[10 Marks]

Solution 5:

- i) If collateralized, swap will have exposure backed by risk free assets via a process of margins. There are initial margin and variation margins and hence nothing will be left in terms of counterparty credit risk. Collateralized derivatives are considered to be less risky, because such transactions pose less counterparty risk, given that the collateral can be used to offset any credit losses.

Collateralized derivatives are much safer instruments than their non-collateralized counterparts, and they are usually valued at risk-free rates.

Types of collateral

- Cash
- Government securities (often direct obligations of G10 countries: Belgium, Canada, France, Germany, Italy, Japan, Netherlands, Sweden, Switzerland, United Kingdom and the United States)
- Mortgage-backed securities (MBSs)
- Corporate bonds/commercial papers
- Letters of credit/guarantees
- Equities
- Government agency securities

[3]

- ii) The value of the swap can be calculated by using the cashflow of the excess payment of 1%, there would be 4 payments which would happen from the start of 1 year (i.e. first payment would not be discounted).

Time from Start of transaction	Time at valuation	Payment	Discounted value
0			
1	0	1	1.00
2	1	1	0.96
3	2	1	0.92
4	3	1	0.89
Total			3.78

Total value is Rs 3.78 Cr.

[3]

- iii) As a first step we would estimate the average default rate from the information

	Number of swaps	Defaults on payment	
2015	890	43	4.83%
2016	740	24	3.24%
2017	1010	71	7.03%
2018	780	28	3.59%
2019	790	67	8.48%

**average
default 5.44%**

Using the default rate estimate the expected loss from the uncollateralised swap

Time	Payment	Expected loss	Recovery	Expected principal at end	Payment	Discounted Value
0	1.0000	0.0000	0.0000	1.0000	1.0000	1.0000
1	1.0000	0.0544	0.0163	0.9456	0.9620	0.9250
2	0.9456	0.0514	0.0154	0.8943	0.9097	0.8410
3	0.8943	0.0486	0.0146	0.8457	0.8602	0.7647
					Total	3.5307

New value of swap allowing for default 3.53 Cr

Capital requirement is 300% of the expected loss

Initial value	3.78
Expected value	3.53
Expected loss	0.25
Capital required	$3 \times 0.25 = 0.75$

[8]

- iv) Use market data which include all uncollateralised swaps would ignore the underlying strength and weakness of the company (Broch) and would consider overall market experience. The swap are very common with wide variety of lenders which may be different from road building company. Thus overall the defaults can be much lower or much higher compared with expected strength of Broch Ltd. Using the corporate bond approach would assess the default risk associated with exposure to Broch Ltd.

This approach would ignore risk associated with the variability of the underlying interest rate risk with and only provide the default probability rather than loss given default for credit risk exposure.

The residual risk would be interest movement – this can be accounted by considering loss given default derived from collateralised swap with is market traded and probability of default from the corporate bond.

Total risk = loss given default x probability of default.

[5]

[19 Marks]

Solution 6:

- i) The hedge ratio is the ratio of the size of the position taken in futures contracts to the size of the exposure in the underlying asset. Hedgers seek to minimise the variance of the change in the value of their combined position. If there is no cross-hedging, then a hedge ratio of 1 would be sensible. However, this is not optimal when cross-hedging.
The minimum variance hedge ratio (or optimal hedge ratio) is the ratio of futures position relative to the spot position that minimizes the variance of the position

Minimum Variance Hedge ratio = $\text{correlation} \times (\text{Standard deviation of spot price returns} / \text{Standard deviation of future price returns})$

[2]

- ii)

Spot	4
Future	6
Correlation	0.75
Hedge ratio	0.5

With hedge ratio of 0.5 the oil consumer should only purchase $10000 \times 0.5 = 5000$ barrels hence 5 contracts.

[2]

- iii) Key risks

- Liquidity risk: Need to constantly balance the Margin account
- The above calculation have inherent limitation both model as well as data – The modelling approach of using Black-Scholes may not be appropriate. The volatilities can vary and may not follow the historical path.
- In case his future consumption required are lower or higher he maybe under or over hedged.

[2]

[6 Marks]**Solution 7:**

- i) Given we are expecting the yields of government bonds to fall the value of the bonds is expected to increase hence we should aim to maximize the allocation to the government bonds. In similar manner , if we expect higher default then aim should be to minimize the exposure. Thus if company aims to have maximum allowable equity exposure it should look at bond exposure of 10%

	Current	Future (Guidance)	Expected
Bonds Gov	50%	Min 25%	40%
Bonds Corp	30%	Max 25% and min 10%	10%
Indian equity	10%	30%	30%
US equity	10%	20%	20%

[2]

- ii) Advantages of using future

- Liquidity constraints might inhibit direct sale without significant impact on the prices.
- Higher liquidity of the future contracts may help to carry out the exposure shift at faster pace
- The transaction cost for the future contracts are generally lower compared to direct sale.
- There can be possible taxation benefits associated.
- We can vary the exposure based and time the transaction based on the insights of the fund manager.

Challenges

- Given the multicurrency transaction there may involve multiple layers of future to get the right exposure
- The market for corporate bonds future would be limited hence overall would require selling assets and reinvesting the money received for the future equity transactions.
- There can be possible basis risk created for the corporate bond futures
- There can be duration mismatch for the government bond futures

[4]

- iii) Asset allocation should be as at the date in future which is 6 months with the assets of 1000cr the follow would be true.

	Current		Future		Difference
Bonds Gov	50%	500.00	40%	400.00	- 100.00
Bonds Corp	30%	300.00	10%	100.00	- 200.00
Indian equity	10%	100.00	30%	300.00	200.00
US equity	10%	100.00	20%	200.00	100.00

Given there are no future available for the corporate bonds, we would have to sell them in the market which would result in excess cash being received by the company. Here we would need to consider 6 months risk free exposure for investments.

Future price is calculated by arbitrage free concept $=S(0) * \exp(rt)$. we aim to go long in Indian and US equities and short on Gov bonds. As mentioned above we would be selling the bonds and investing the amount in Risk free assets. The below table gives other transactions

Interest rate	3.00%	Index S&P	3000	US	1%	
USD	75	Index BSE	35000	Cr	10,000,000.00	
	Amount(A)	$B = A * CR / FX$	Price of index	Interest 6 monthly	Price of future ($\exp(r) * S(0)$)	Number of Future (B/Future P)
India (BSE)	200.00	2,000,000,000	35000	3%	36066	55,454
US equity	100.00	13,333,333	3000	1%	3030	4,400

Government bonds

Interest rate	3.00%					
					10,000,000.00	
	Amount(A)	$B = A * CR / FX$	Price of index	Interest 6 monthly - Coupon Payment	Price of future ($\exp(r) * S(0)$)	Number of Future (B/Future P) - ive represent sell the futures
G Bonds	- 100.00	-1,000,000,000	10,000,000.00	-3%	9704455	- 103

[6]

- iv) The value of bond which pays perpetually and priced at Par with yield [y] can be stated as below
 $Price = P(\frac{y}{(1+y)^1} + \frac{y}{(1+y)^2} + \frac{y}{(1+y)^3} \dots)$

Which is equal to :

$$P(y) = 1/y$$

Where:

Y=annual effective yield First and second derivative of the price

First differential

$$P(y)' = -1/y^2$$

Second differential

$$P(y)'' = 2(1/y^3)$$

$$Convexity = P(y)'' / P(y) = 2/y^2$$

[5]

[17 Marks]

Solution 8:

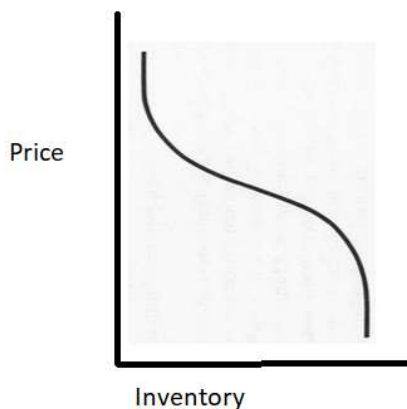
- i) Black Scholes pricing formula assumes that prices cannot be negative. While this may be true in the case of stocks the event has proved that this need not be the case for commodities with non-trivial storage costs. Hence the pricing models must be made more flexible.

The following approaches can be adopted:

- The volatility, correlations etc. depend on historical returns and not prices. So, the pricing model maybe switched to an assumption of normal distribution for prices below a particular pre-defined threshold
- So whenever the price of the commodity approaches/breaches the threshold an automatic switch can be put in place that changes the pricing model from one that assumes lognormal distribution of prices to one that assumes normal distribution of prices.
- The models/threshold maybe aligned with that used by exchanges
- When roll overs are to be modelled modelling the spread rather than yield may make more sense when similar scenarios arise in the future

[5]

- ii) The typical curve of price vs inventory is expected to be of an inverted S structure. The price rises steeply when the inventory is less than a particular low-level implying demand significantly higher than supply. The price also falls steeply beyond a particular high level of inventory with when supply outstrips demand. But typically the price curve should be ending at zero when inventory levels are very high. Now under the new circumstances the model may have to be revised to consider prices below zero as well.



(Diagram is provided only for understanding and not needed for the examination)

[3]

[8 Marks]
